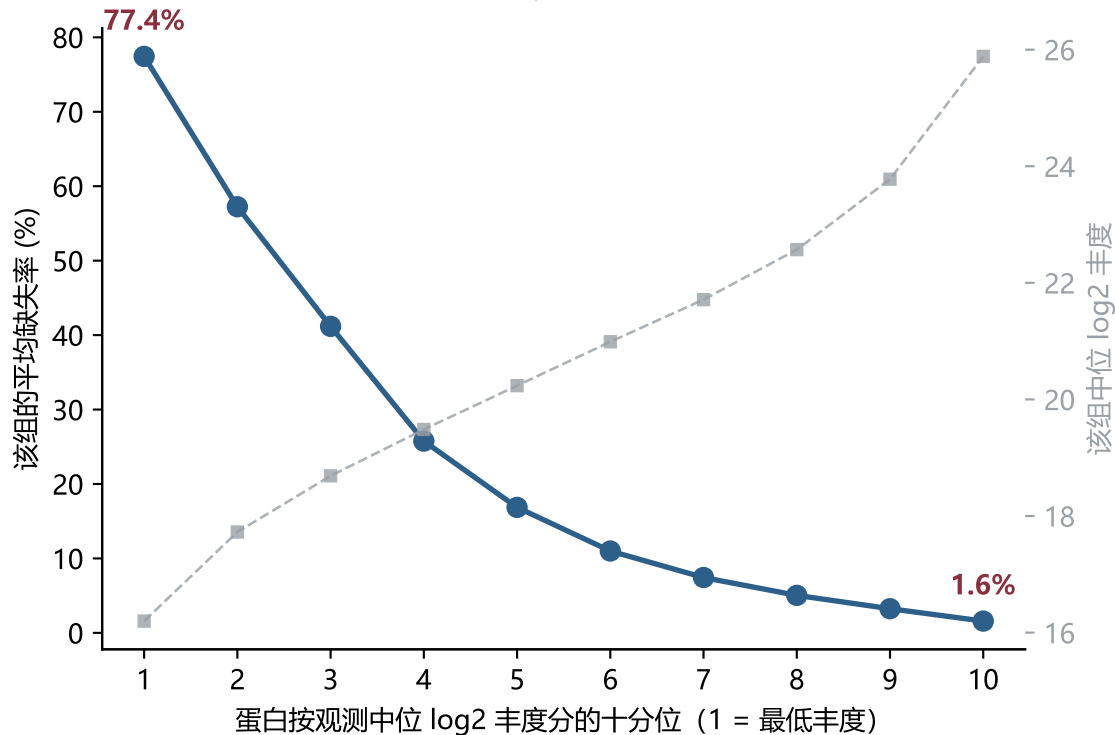


缺失以丰度依赖的检出限为主 (MNAR)
最低丰度组缺 77%，最高丰度组只缺 1.6%



Spearman(中位丰度, 缺失率) = -0.859; 缺失指示回归中仅丰度代理即达伪 R² 0.351, 技术+生物因子仅 0.004。数据: scripts/audit/diagnose_missing.py